

Configuration channel pins.

CC1/CC2/CC. A CC signal is used to detect connection, orientation and also configure the devices into their operating modes. Based on connection orientation, one of the CC lines acts as V_{CONN} signal to provide power to the other end.

Power pins. V_{BUS} are for USB bus power and can carry up to 20V/5A. These are available only after connection detection.

V_{CONN} are used to provide power to the attached plug and are applied via the unused CC pin based on connection orientation.

Auxiliary pins. SBU1/SBU2.

The side-band use (SBU) pins are provided for adding additional functionality either through future USB specification or during alternate mode functionality or as an audio accessory.

USB 2.0 pins. DP/DM. D+ and D- pins are used to support USB 2.0 in all three modes, namely, low speed, full speed and high speed. These pins are provided as pairs in a receptacle to support the plug-flip-ping feature and are shorted in the plug connector.

Super-speed pins. Rx_n +/ Rx_n -, Tx_n +/ Tx_n -. These lines are used to implement Rx and Tx of USB 3.1 super-speed interface. These are shared by other interfaces when in alternate mode.

USB PD specification

Even before the new USB Type-C connector was introduced, USB-IF published USB PD specification, which defines how a USB interface can deliver 20V to its port partner. Version 1.0 of the specification uses BFSK on V_{BUS} line for protocol communication.

Later, with the introduction of USB Type-C interface connector, USB PD specification has moved on from BFSK on V_{BUS} to BMC based protocol communication on the CC signal line. A new specification version 2.0 has been released with the

above-mentioned changes allowing USB PD protocol to negotiate higher power with the port partner and also jump to alternate functionalities like Displayport, MHL and others over USB Type-C signals.

A USB Type-C device can be broadly classified into two categories: USB Type-C device and USB Type-C device with PD capability. A standalone USB Type-C device supports up to 5V/3A operations and orientation detection, whereas a USB-Type-C device with PD capability supports up to 20V/5A opera-

capabilities. Also, a DFP can identify if a UFP is connected directly or through a USB Type-C cable with V_{CONN} requirement.

USB Type-C host (DFP) to USB Type-C device (UFP) with PD support. A USB Type-C device with PD support is an advanced USB Type-C device that provides or consumes up to 20V/5A power. It can also enter modal operations to reuse USB Type-C signals for additional functionalities. Fig. 4 illustrates USB Type-C device with PD support connection.

PD communication using PD controller starts after a successful USB Type-C connection. At the start of the process, a PD source starts sending its capabilities over the CC line using BMC encoding. The sink decodes capabilities of the source and requests the power it requires. Based on the request, the source may accept the request and change its V_{BUS} to output the requested power. The source then sends a source-ready packet to the sink indicating acceptance.

When the port partners agree on power, state is called contract. This is the basic state requirement for any further protocol or mode negotiation.

Things you can do with USB Type-C

A USB Type-C cable plays a key role in the USB Type-C ecosystem and can dictate the behaviour of the ecosystem. In the USB Type-C environment, cable products are called electronically-marked cables that can be built with intelligence for robust and better user experience.

The flip side to this is that users need to be educated about the cables' capability. This section explores key things that a USB Type-C cable can do, which the legacy USB cables can not.

Can talk. Unlike legacy USB cables, USB Type-C cables can be electronically marked and these

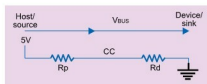


Fig. 3: USB Type-C DFP-UFP connection

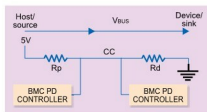


Fig. 4: USB PD DFP-UFP connection

tion and multiple interfaces using alternate modes.

USB Type-C host or down stream facing port (DFP) to USB Type-C device or upstream facing port (UFP) connection. When two simple forms of USB Type-C devices, a USB host and a USB device, are connected, a USB host will present an R_p on the CC line and a USB device will present an R_d on the CC line as shown in Fig. 3.

When the above devices get connected, a drop in the CC line for a DFP and a rise in the CC line for a UFP confirms a connection. The rise and drop is to a predefined voltage range as defined by USB Type-C specification.

Based on the CC line voltage range, a UFP will identify the type of DFP connected and its power